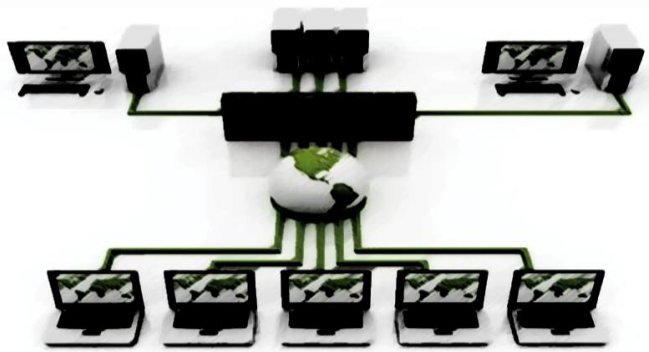


ARBAMINCH UNIVERSITY



ARBAMINCH INSTITUTE OF TECHNOLOGY

FACULTY OF COMPUTING AND SOFTWARE ENGINEERING



FUNDAMENTALS OF NETWORKING [SEng7022]

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Data Communication

Information is defined as a **collection of facts** from which conclusions may be drawn is an important resource. The need of information has increased from time to time leads to the need of sharing of information among different agents. **Data communication** is the exchange of information between two agents. In the **Old paradigm** A single powerful computer serving all the needs of an organization i.e. **Sneaker-net** a Method of sharing data by copying it to a disk and carrying it from computer to computer. On the contrary of the **New paradigm Computer networks** is a large number of separate (autonomous) but internetworked (being able to exchange information) computers doing the job. **A computer network** is an interconnected collection of autonomous computers. **Interconnected** means two computers have the ability to exchange information using some transmission media. **Autonomous** refers where no one computer controls any other computer (i.e. no computer can forcibly start or stop another computer). **Computers** can be PC's, workstations and other "specialized" computers such as hubs, switches and routers, and they can be geographically located anywhere.

Components of Data Communication System

Data communication system has 5 components.

Message: the information to be communicated (text, numbers, pictures, sound, video - or combinations)

Sender: it is the device, computer, video camera, etc.

Receiver: still the device.

Medium: the physical **path** by which a message travels from sender to receiver.

Protocol: the set of **rules** that govern data communications; an **agreement** between the communicating devices.

Network Categories

- ✓ Local Area Network (LAN): usually privately owned and links devices in a single office, building or campus.
- ✓ Wide Area Network (WAN): covering large geographic area; may utilize public, leased, or private communications equipment
- ✓ Metropolitan Area Network (MAN): designed to extend over an entire city; it may be a single network or interconnected LANs
- ✓ Personal Area Network (PAN): meant for one person. e.g. a wireless network connecting a computer with its mouse, keyboard and printer.

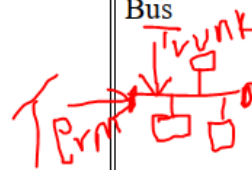
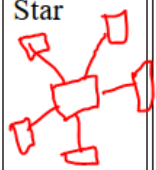
Network Architecture

Includes the type of computers on the network and determines how network resources are handled. The two common types of network architecture are **Peer to Peer** and **Client to Server**. In the case of the peer to peer each node considered as equal in terms of resource sharing and responsibilities. But in the case of the client server the client request for a service and the server deliver the resource (resourceful).

Network Topology

Network Topology refers to the way in which multiple devices are interconnected via communication links. There are two types of topology i.e. the physical and logical. **Physical Topology** Refers to the way in which a network is laid out physically. It refers to the arrangement or physical layout of computers, cables, and other components on the network. **Physical Topology** Can be referred as Physical layout, Design, Diagram, Map and in this topology two or more devices connect to a link; two or more links form a topology. **Logical topology** is bound to network protocols and describe how data is moved across the network. The choice of one topology over another will have an impact on: The types of equipment that the network needs, the growth of the network – scalability, the way the network managed and etc. four well known network

topologies are possible for the implementation of a given network design. These are the **bus, star, ring, mesh, and hybrid network topology**.

Topology	Advantages	Disadvantages
	- Use of cable is economical. - Media is inexpensive and easy to work with. - System is simple and reliable. - Bus is easy to extend.	- Network can slow down in heavy traffic. - Problems are difficult to isolate. - Cable break can affect many users.
	- System provides equal access for all computers. - Performance is even despite many users.	- Failure of one computer can affect the rest of the network. - Problems are hard to isolate. - Network reconfiguration disrupts operation.
	- Modifying system and adding new computers is easy. - Failure of one computer does not affect the rest of the network.	- If the centralized point fails, the network fails.
	- System provides increased redundancy and reliability as well as ease of troubleshooting.	- System is expensive to install because it uses a lot of cabling.

$$L = \frac{N(N-1)}{2}$$

Modes of transmission

Refers to the direction of signal flow between two linked devices. It can be **Simplex**: unidirectional, only one of the devices can transmit E.g. TV transmission, radio, **Half-duplex**: both can transmit and receive, but not at the same time E.g. wireless handset (walkie-talkie) or **Full-duplex**: both can transmit and receive at the same time E.g. Telephone transmission.

Transmission media

It is physical media that carries a signal from the transmitter to the receiver. The information or signal transmitted from one device to another is through electromagnetic signals. **Electromagnetic signals** include power, voice, radio, waves, infrared light, visible light, ultraviolet light, X-rays, and gamma rays. **Guided** transmission media: uses a cabling system that guides the signals along a specific path. E.g. Fiber Optics, Twisted Pair, Coaxial Cable etc. **Unguided**: consists of a means for the data signals to travel but nothing to guide them along a specific path - wireless. E.g., Radio wave, Satellite, etc.

Uses of Computer Networks

- ✓ for resource sharing including programs, equipment, data (mostly databases on central servers)
- ✓ a communication medium – e-mail, writing a report together by making changes on an online document
- ✓ videoconferencing – to hold meetings by hearing and seeing each other
- ✓ electronic business
 - business to business - placing orders, ...
 - business with consumers, usually called e-commerce – home shopping

Transmission Impairments

Noise: additional signals inserted between transmitter and receiver is known as noise. Noises can be caused by Thermal: Due to thermal worry of electrons in a conductor, Crosstalk: A signal from one line is picked up by another, Impulse Noise : is sharp quick spikes on the signal caused from electromagnetic interference, lightning, sudden power switching, electromechanical switching, etc and inter-modulation noise means a noise may occurred during changing from digital to analog or from analog to digital.

Attenuation: it is a phenomenon which occurs when signal strength falls off with distance. It depends on medium, such as Fiber optic cables carries signal without attenuation up to 2Km. In communicating entities, received signal strength must be enough to be detected and again it must be sufficiently higher than noise to be received without error.

Delay Distortion: this distortion can only occur in guided media. Propagation **velocity varies** with frequency. Various frequency components of a signal will arrive at the receiver at different times.

Multiplexing

It is a term used to refer to a process where **multiple** analog message signals or digital data streams are combined **into one signal** over a shared medium to share an expensive resource. Multiplexing divides the capacity of the low-level communication channel into several higher-level logical channels, one for each message signal or data stream to be transferred. A reverse process, known as **demultiplexing**, can extract the original channels on the receiver side. The two most basic forms of multiplexing are :

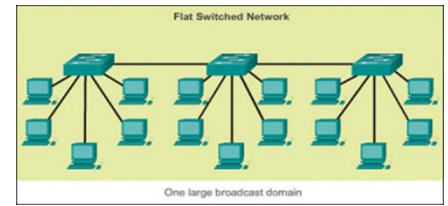
- ✓ Time-division multiplexing (TDM) –for digital signal
- ✓ Frequency-division multiplexing (FDM) –for analog signal

Chapter Two: Data Communication Layer

Network architecture is a design of communication networks for specification of a network's physical components and their functional organization and configuration, its operation principles and procedures as well as data formats use. Network architecture is predominantly expressed by its use of the internet protocol suits. Network Line Configuration defines the attachment of communication devices to a link. Both point to point and multipoint connection are line configuration to connect two or more devices in link. Link is a physical communication pathway that transfers data from one device to another. A point-to-point line is a line configuration where two and only two devices are connected by a dedicated link. It is unicast connection because there is dedicated link between the two devices. **Multipoint link** is a line configuration in which more than two specific devices share a single link.

Network Design Models

There are basically have 2 network design models. The first one is called Flat model. it is single collision domain and not hierarchical divided. It has some advantage like Easy to design, Easy to implement, it good for small network, and it is easy to maintain and diagnosis. **Its disadvantage includes** Poor security, No redundancy, less speed, No scalability and there are no networks layers.

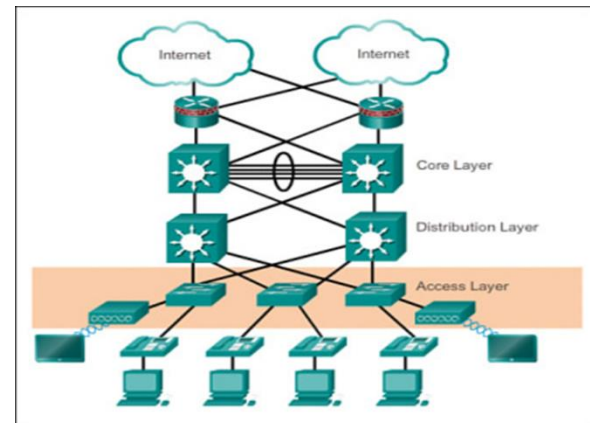


Flat switch network

The Second design model is called **Hierarchical Model**. It is a network model that has **three** network layers or divisions and each layers has their own specific functions.

Thus the model has three layers called

- I. **Access layer:** used for service available port security on layer 2 switch.
- II. **Distribution layer:** aggregate data received from access layer and switches from sending to **core layer**, redundancy and load balancing, control broadcast domain.
- III. **Core layer control:** it is a **network backbone**, it is fast speed layer it connect multiple campus network and data center. Such kind of network design is Difficult to design and Difficult to implement



Hierarchical Model

Network Protocols and Standards

A protocol is a set of rules that governs data communications. A protocol defines what is communicated, how it is communicated, and when it is communicated. for instance, for one computer to send a message to another computer, the first computer must perform the following general steps

- ✓ break the data into small sections called packets
- ✓ add addressing information to the packets identifying the destination computer
- ✓ deliver the data to the network card for transmission over the network

The receiving computer must perform the same steps, but in reverse order

- ✓ accept the data from the NIC
- ✓ remove transmitting information that was added by the transmitting computer
- ✓ reassemble the packets of data in to the original message

The key elements of a protocol are: syntax: refers to the structure or format of the data, semantics: refers to the meaning of each section of bits and timing: refers to when data should be sent and how fast they can be sent.

What are the principal motivation for the development of the OSI model?

The principal motivation for the development of the OSI model is to provide a framework for standardization. Within the model, one or more protocol standards can be developed at each layer. Again **Changes in** standards in one layer **need not affect** already existing software in another layer. Here The **design principle of information hiding** is used:

- ✓ **Lower layers** are concerned with **greater levels of detail**;
- ✓ **Upper layers** are independent of these details.

For two computers, in a computer network, to communicate must talk the same language. Follow the same protocol. A **network protocol** is a formal set of rules, conventions and data structure that governs how computers and other network devices exchange information over a network. **In earlier days**, many of the networks that were built used different hardware and software implementations

They were **incompatible** and it became **difficult for networks** using different specifications to communicate with each other. **The layers are separated from each other by boundaries** called **interfaces**. All requests are passed from one layer, **through the interface**, to the next layer.

Why layering?

Layering reduces complexity (one big problem to smaller ones). It standardizes interfaces (between layers) facilitates modular engineering (different teams work on different modules) and **assures interoperable technology**. Layering also accelerates evolution of networking technology.

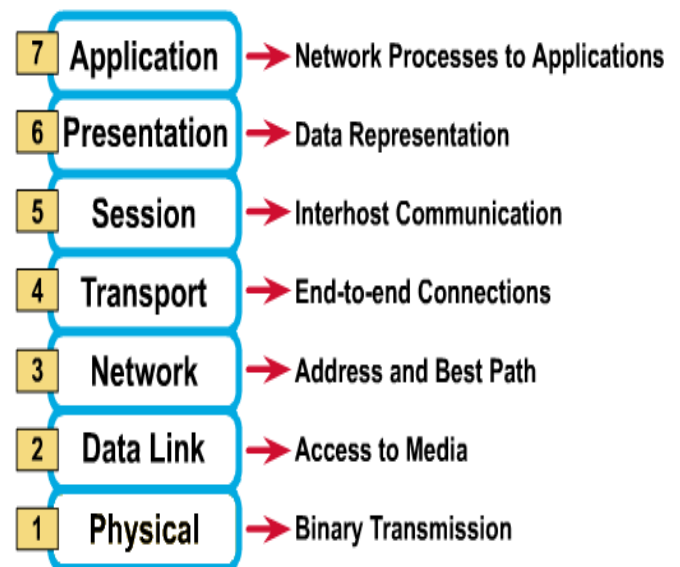
OSI (open system interconnection) Reference Model

It was developed by the International Organization for Standardization (ISO) in 1970. /OSI=open system interconnection. It is a model for a computer protocol architecture and as a framework for developing protocol standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It is now considered the primary Architectural model for inter-computer communications. OSI is a descriptive network scheme. It ensures greater compatibility and interoperability between various types of network technologies. The model Comprises of seven layers and Defines how each layer communicates and works with the layers immediately above and below it.

OSI is important because it brakes Network communication into smaller, more manageable parts. And also allows different types of network hardware and software to communicate with each other. In OSI All layers are independent and changes does not affect other layers.

The Seven OSI Reference Model Layers

Each layer provides a service to the layer above it in the protocol specification. Each layer communicates with the same layer's software or hardware on other computers. The lower 4 layers (transport, network, data link and physical: Layers 4, 3, 2, and 1) are concerned with the flow of data from end to end through the network. The upper three layers of the OSI model (application, presentation and session: Layers 7, 6 and 5) are orientated more toward services to the applications. Data is encapsulated with the necessary protocol information as it moves down the layers before network transit.



Each layer should perform a well-defined function. A layer should be created where a different abstraction is needed. The function of each layer should be chosen with an eye toward defining internationally standardized protocols. The layer boundaries should be chosen to minimize the information flow across the interfaces. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

Layer 7: Application

- ✓ The application layer is the OSI layer that is **closest to the user**.
- ✓ It provides network services to the user's applications.
- ✓ contains all the **higher level protocols** that are commonly needed by users

Layer 6: Presentation

- ✓ The presentation layer ensures that the information that the application layer of one system sends out is **readable by the application** layer of another system.
- ✓ If necessary, the presentation layer translates between **multiple data formats** by using a common format.
- ✓ Provides encryption and compression of data.
- ✓ Examples :- JPEG, MPEG, ASCII, EBCDIC, HTML

Layer 5: Session

- ✓ The session layer defines how to start, control and end conversations (called sessions) between applications.
- ✓ This includes the control and management of multiple bi-directional messages using dialogue control.
 - keeping track of whose turn it is to transmit
- ✓ It also synchronizes dialogue between two hosts' presentation layers and manages their data exchange.
 - preventing two parties from attempting the same critical operation at the same time
- ✓ The session layer offers provisions for efficient data transfer.
 - Check pointing long transmissions to allow them to continue from where they were after a crash
- ✓ Examples: - SQL, ASP (AppleTalk Session Protocol).

Layer 4: Transport

- ✓ It accepts data from above, splits it up into smaller units if need be, passes them to the network layer, and ensure that the pieces all arrive correctly at the other end
- ✓ The transport layer segments data from the sending host's system and reassembles the data into a data stream on the receiving host's system.
- ✓ End-to-end error free delivery of entire message
- ✓ Services include:
 - Service port addressing : Port number
 - Segmentation /reassembly
 - Connection control: Connectionless or connection oriented
 - Flow and error control

Layer 3: Network

- ✓ Defines end-to-end delivery of packets.
- ✓ Defines logical addressing so that any endpoint can be identified.
- ✓ Defines how routing works and how routes are learned so that the packets can be delivered.
- ✓ The network layer also defines how to fragment a packet into smaller packets to accommodate different media.
- ✓ Examples: - IP, IPX, AppleTalk.

Layer 2: Data Link

- ✓ The data link layer provides access to the networking media and physical transmission across the media and this enables the data to locate its intended destination on a network.
- ✓ The data link layer provides reliable transit of data across a physical link by using the Media Access Control (MAC) addresses.
- ✓ The data link layer uses the MAC address to define a hardware or data link address in order for multiple stations to share the same medium and still uniquely identify each other.
- ✓ Concerned with network topology, network access, error notification, ordered delivery of frames, and flow control.
- ✓ Examples: - Ethernet, Frame Relay, FDDI.

Layer 1: Physical

- ✓ The layer that actually interacts with the transmission media
- ✓ the physical part of the network that connects network components together
- ✓ involved in physically carrying information from one node in the network to the next
- ✓ The physical layer deals with the physical characteristics of the transmission medium.
- ✓ It defines
 - mechanical: the size and shape of the network connector, how many pins does the network connector has and what each pin is used for
 - electrical: how many volts represent a 1 and how many a 0
 - timing: how many nanoseconds a bit lasts
 - Whether communication is one way or in both directions simultaneously
- ✓ Such characteristics as voltage levels, timing of voltage changes, physical data rates, maximum transmission distances, physical connectors, and other similar attributes are defined by physical layer specifications.
- ✓ Examples :- EIA/TIA-232, RJ45, BNC

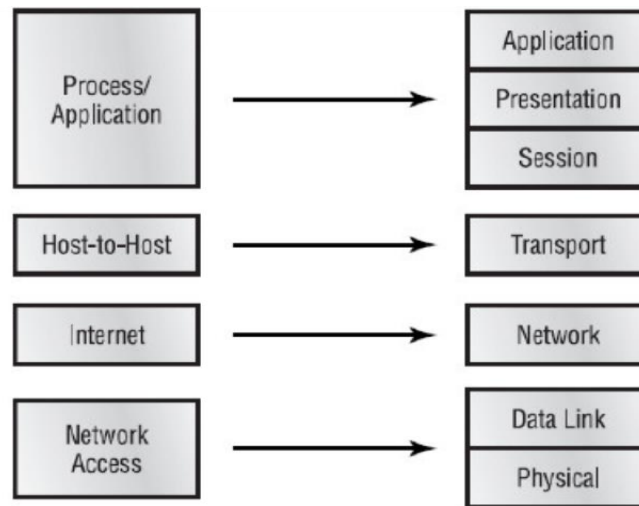
Summary of OSI model

There was **no standard for networks** in the early days and as a result it was difficult for networks to communicate with each other. The International Organization for Standardization (ISO) recognized this. and researched various network schemes, and in 1970 introduced the Open Systems Interconnection (OSI) reference model. **The OSI reference model** has standards which ensure vendors (sellers) greater compatibility and interoperability between various types of network technologies. The OSI reference model organizes network functions into **seven numbered layers**. Each layer provides a service to the layer above it in the protocol specification and communicates with the same layer's software or hardware on other computers. Layers 1-4 are concerned with the **flow of data from end -to- end** through the network and Layers 5-7 are concerned **with services to the applications**.

TCP/IP Reference Model

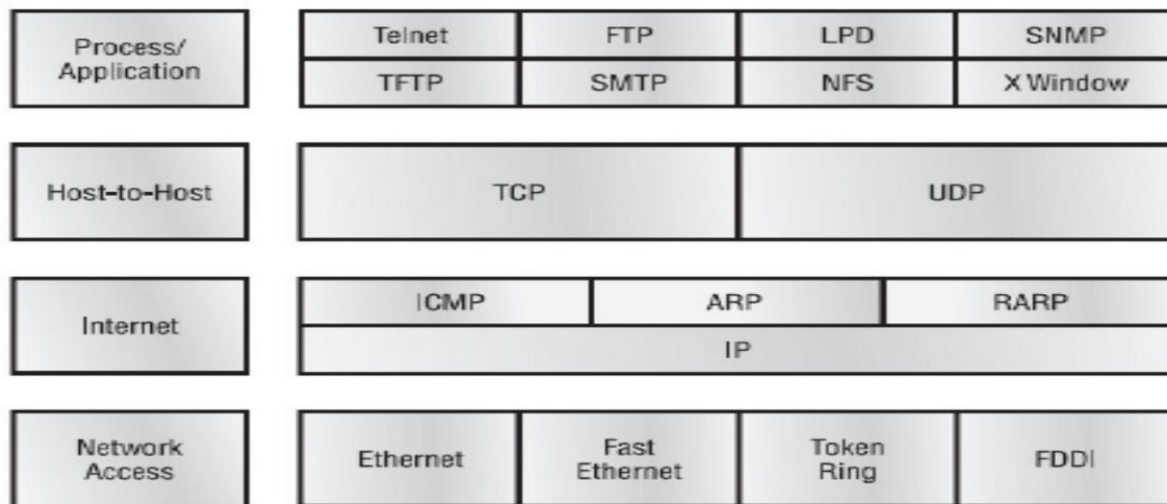
Used in the grandparent of all wide area computer Networks, the ARPANET. Developed by Department of Defense (DoD) to ensure and preserve data integrity as well as maintain communication in the event of Catastrophic war. Condensed version of OSI model contains four layers instead of seven **The Process/Application** protocols for **node-to-node application** communication and also controls user-interface specifications. **The Host-to-Host** parallels the functions of the OSI's Transport layer, defining protocols for setting up the level of transmission service for applications. **The Internet layer** corresponds to the OSI's Network layer, designating

the protocols relating to the logical transmission of packets over the entire network. **Network Access layer** is the equivalent of the **Data Link and Physical layers** of the OSI model. It oversees hardware addressing and defines protocols for the physical transmission of data.



The model **does not clearly distinguish the concepts of service, interface, and protocol**. The TCP/IP model **is not at all general** and **is poorly suited to describing** any protocol stack other than TCP/IP. The host-to-network layer is not really a layer at all. The model **does not** distinguish (or even mention) the physical and data link layers.

Protocols



Application/Process layer

Telnet - allows a user on a remote client machine, called the Telnet client, to access the resources of another machine, the Telnet server. Telnet makes client machine appear as though it were a terminal directly attached to the server.

File Transfer Protocol (FTP) - is the protocol that actually lets us transfer files, and it can accomplish this between any two machines using it. Usually users are subjected to authentication.

Network File System (NFS) - a protocol specializing in file sharing allowing two different types of file systems to interoperate.

Simple Mail Transfer Protocol (SMTP) - uses a spooled, or queued, method of mail delivery. POP3 is used to receive mail.

Simple Network Management Protocol (SNMP) - collects and manipulates valuable network information. This protocol stands as a watchdog over the network, quickly notifying managers of any sudden turn of events.

Domain Name Service (DNS) – resolves hostnames—specifically, Internet names, such as www.amu.edu.et to the IP address 10.6.10.3

Dynamic Host Configuration Protocol (DHCP) - gives IP addresses to hosts. It allows easier administration and works well in small-to-even-very large network environments.

Host-to-Host layer

Transmission Control Protocol (TCP) - takes large blocks of information from an application and breaks them into segments. It numbers and sequences each segment so that the destination's TCP protocol can put the segments back into the order the application intended. Uses three way handshaking

User Datagram Protocol (UDP) - does not sequence the segments and does not care in which order the segments arrive at the destination. But after that, UDP sends the segments off and forgets about them. It doesn't follow through, check up on them, or even allow for an acknowledgment of safe arrival — complete abandonment. TCP for reliability and UDP for faster transfers.

TCP and UDP must use port numbers to communicate with the upper layers, because they're what keeps track of different conversations crossing the network simultaneously. These port numbers identify the source and destination application or process in the TCP segment. **There are 216 = 65536** ports available. Well-known ports range from 0 to 1023. Registered ports range from 1024 to 49151. Registered ports are used by applications or services that need to have consistent port assignments. Dynamic or private ports range from 49152 to 65535. These ports are not assigned to any protocol or service in particular and can be used for any service or application.

PORT NUMBER	TRANSPORT PROTOCOL	SERVICE NAME	RFC
20, 21	TCP	File Transfer Protocol (FTP)	RFC 959
22	TCP and UDP	Secure Shell (SSH)	RFC 4250-4256
23	TCP	Telnet	RFC 854
25	TCP	Simple Mail Transfer Protocol (SMTP)	RFC 5321
53	TCP and UDP	Domain Name Server (DNS)	RFC 1034-1035
67, 68	UDP	Dynamic Host Configuration Protocol (DHCP)	RFC 2131
69	UDP	Trivial File Transfer Protocol (TFTP)	RFC 1350
80	TCP	HyperText Transfer Protocol (HTTP)	RFC 2616
110	TCP	Post Office Protocol (POP3)	RFC 1939
119	TCP	Network News Transport Protocol (NNTP)	RFC 8977
123	UDP	Network Time Protocol (NTP)	RFC 5905
135-139	TCP and UDP	NetBIOS	RFC 1001-1002
143	TCP and UDP	Internet Message Access Protocol (IMAP4)	RFC 3501
161, 162	TCP and UDP	Simple Network Management Protocol (SNMP)	RFC 1901-1908, 3411-3418
179	TCP	Border Gateway Protocol (BGP)	RFC 4271
389	TCP and UDP	Lightweight Directory Access Protocol	RFC 4510
443	TCP and UDP	HTTP with Secure Sockets Layer (SSL)	RFC 2818
500	UDP	Internet Security Association and Key Management Protocol (ISAKMP) / Internet Key Exchange (IKE)	RFC 2408 - 2409
636	TCP and UDP	Lightweight Directory Access Protocol over TLS/SSL (LDAPS)	RFC 4513
989/990	TCP	FTP over TLS/SSL	RFC 4217

<https://ipwithease.com>

If a port is closed/blocked, you cannot communicate with the computer by the protocol using that port. Eg. If port 25 is blocked you cannot send mail. Firewalls by default block all ports.

Internet Layer

Internet Protocol (IP) essentially is the Internet layer. The other protocols found here merely exist to support it. It can do this because all the machines on the network have a software, or logical, address called an IP address.

Internet Control Message Protocol (ICMP) works at the Network layer and is used by IP for many different services. ICMP is a management protocol and messaging service provider for IP. The following are some common events and messages that ICMP relates to:

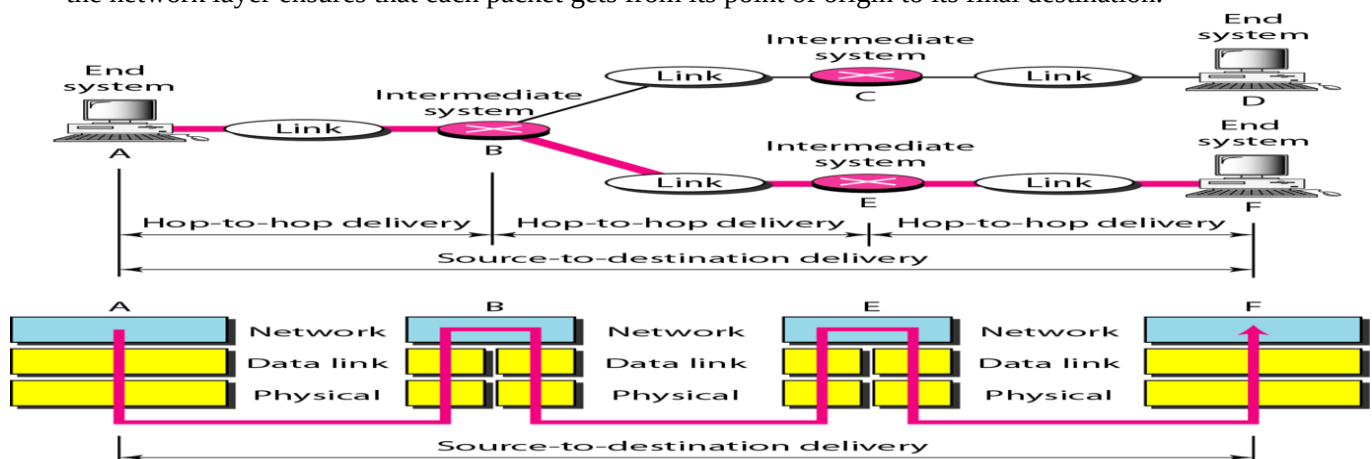
- ✓ **Destination Unreachable** If a router can't send an IP datagram any further, it uses ICMP to send a message back to the sender, advising it of the situation.
- ✓ **Buffer Full** If a router's memory buffer for receiving incoming datagrams is full, it will use ICMP to send out this message until the congestion abates.
- ✓ **Hops** Each IP datagram is allotted a certain number of routers, **called hops**, to pass through. If it reaches its limit of hops before arriving at its destination, the last router to receive that datagram deletes it.
- ✓ **The executioner router then uses ICMP** to send a message, informing the sending machine of the end of its datagram.
- ✓ **Ping** (Packet Internet Groper) uses ICMP echo messages to check the physical and logical connectivity of machines on a network.
- ✓ **Trace route** Using ICMP timeouts, Traceroute is used to discover the path a packet takes as it traverses an internetwork.

Address Resolution Protocol (ARP) finds the hardware address of a host from a known IP address. ARP interrogates the local network by sending out a broadcast asking the machine with the specified IP address to reply with its hardware address

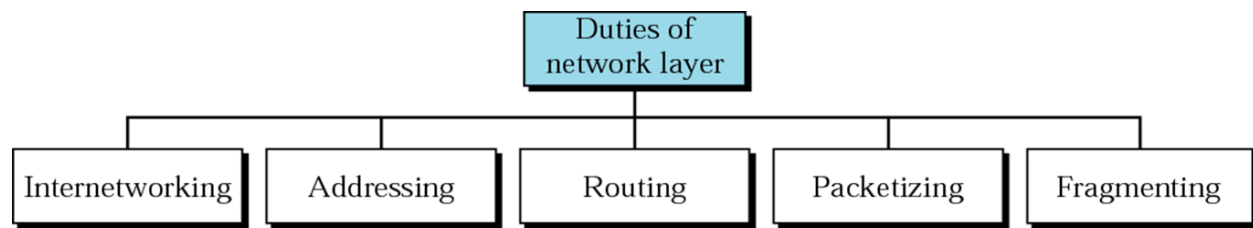
Reverse Address Resolution Protocol (RARP) discovers the identity of the IP address for diskless machines by sending out a packet that includes its MAC address and a request for the IP address assigned to that MAC address. A designated machine, called a RARP server, responds with the answer, and the identity crisis is over.

Chapter Three: Network Layer

Network Layer (3rd OSI Layer) Concerned with getting packets from source to destination. The network layer must know the topology of the subnet and choose appropriate paths through it. When source and destination are in different networks, the network layer must deal with these differences. The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks. Whereas the data link layer oversees the delivery of the packet between two systems on the same network, the network layer ensures that each packet gets from its point of origin to its final destination.



If two systems are connected to the same local network, there is usually no need for a network layer. However, if the two systems are attached to different networks with connecting devices between the networks, there is often a need for the network layer to accomplish source-to-destination delivery.



Network Layer Protocols

IPv4 was not designed with this feature in mind, although mobility extensions exist. IPv6 has a better designed solution. Network layer protocols include: IPv4/IPv6, DVMRP (Distance Vector Multicast Routing Protocol), ICMP(Internet Control Message Protocol), IGMP(Internet Group Multicast Protocol), PIM-SM(Protocol Independent Multicast Sparse Mode), PIM-DM(Protocol Independent Multicast Dense Mode), IPsec(Internet Protocol Security), IPX(Internetwork Packet Exchange), RIP(Routing Information Protocol), Interior Gateway routing protocol(IGRP), Enhanced IGRP(EIGRP), Open Shortest Path First(OSP), DDP(Datagram Delivery Protocol), BGP(Border Gateway Protocol),

Routing Basics

The network layer of the OSI model provides routing functions and establishes the route between the source and destination through intermediary nodes (routers). Routing is one of the fundamental aspects of networking. The ability of routers to learn possible routes (rather than make you manually configure and constantly update the routes) is one of the primary reasons that ARPANET which originally connected seven sites, scaled into the modern Internet in only a few short years. A router is connected to at least two networks, and forwards data packets from a network to another. When it receives an incoming packet, it decides the next router (hop) to which the packet should be forwarded towards its destination. A routing table maintains the best routes to various network destinations.

The construction of routing tables is very important for efficient routing, and any corruption of the table may lead to significant routing problems. To create and maintain a routing table, a router communicates with other routers using routing protocols. A routing protocol allows routers to share information about networks and their proximity to each other. The routing protocol running in a router will learn all available paths to a network (based on the information advertised from different routers), but only determine the best path to be stored in the routing table. Whenever the topology of a network changes (e.g., a new link is added or removed), the routing protocol will determine another best path (if needed) and update the routing table accordingly for the routing table in the updated topology. Routing involves two basic activities:

- ✓ determining optimal routing paths and
- ✓ Transporting packets through a network.

Path determination occurs at Layer 3, the network layer. The **path determination function** enables a router to evaluate the available paths to a destination and to **establish the best path**. **Routing services** use network topology information when evaluating network paths. This information can be configured by the network administrator (**static routes**) or collected through **dynamic processes (routing protocols)** running in the network. **To route, a router needs to know:**

- ✓ Destination addresses

- ✓ Sources it can learn from
- ✓ Possible routes
- ✓ Best route

Routed networks are often **large** and **complex**, and it would be excessively **difficult to manage** and **update** network information on all routers all the time. Several **algorithms** have been developed to help address these difficulties. These algorithms allow the routers to learn about the network and then make decisions based on that information. **To learn paths (or routes) through a network, and make decisions on where to send packets, a router use:**

- ✓ Destination address - Typically the Internet Protocol (IP) address of the data's (packet) destination.
- ✓ Source address - Where the information came from (typically an IP address).
- ✓ Possible routes - Routes that can get information from its present location or source to some other location (the destination or closest known point).
- ✓ Best route - The best path to the intended destination. ("Best" can mean many things.)
- ✓ Status of routes - The current state of routes, which routers track to ensure timely delivery of information.

Routers often make decisions about the best possible path to get information from a source to a destination. "Best," however, is loosely defined, and it depends on what is valued by the network. These **measurements of value** are referred to as **metrics**. Several metrics are listed here:

- ✓ Hop count - Number of times a packet goes through a router.
- ✓ Delay time - Time required to reach the destination.
- ✓ Reliability - Bit-error rate of each network link.
- ✓ Maximum transmission unit (MTU) - Maximum message length (or packet size) allowed on the path.
- ✓ Cost - Usually some combination of other metrics.

To forward network traffic to the proper locations, the routers on your network must have the correct entries in their routing tables. To do so routers will use either **static** or **dynamic** routing techniques. With **static routing**, network administrators must **manually** create and modify the routing table entries. **Dynamic routing** uses a **specialized routing protocol** to update the table entries **automatically**.

Static routes use a programmed route that a network administrator enters into the router. A network administrator configures information about remote networks **manually**

Benefits

- ✓ No overhead on the router CPU
- ✓ No bandwidth usage between routers
- ✓ Adds security

Disadvantage

- ✓ Administrator must really understand the internetwork
- ✓ If a network is added to the internetwork, the administrator has to add a route to it on all routers
- ✓ Not feasible in large networks

Dynamic route uses a route that a routing protocol **adjusts automatically for topology or/and traffic changes**. When routers use dynamic routing, this information is learned from other routers.

Benefits

- ✓ scalability and adaptability
- ✓ can grow more quickly and larger
- ✓ adapt to changes in the network topology

Disadvantage

- ✓ complex structure
- ✓ Adds overhead on the router CPU

- ✓ Routers use bandwidth to send routing table updates
- ✓ Security issue

Internet Routing

The Internet comprises a large number of interconnected heterogeneous routing domains called **autonomous system (AS)**. AS is essentially a collection of IP networks and routers under the same administration that share a common routing strategy. An Internet service provider (ISP) is the typical example of an AS. For the outside world, an AS is regarded as a single routing entity. Each AS must have a **16-bit AS number**, which is **unique** and assigned by a regional Internet registry such as the American Registry of Internet Numbers (<http://www.arin.net>). Some routing protocols require the AS number to make routing decisions.

Interior Vs Exterior Routing Protocols

The routing protocols used **within** an AS are called **interior gateway protocols (IGPs)**, and the ones used **between ASs** are called **exterior gateway protocols (EGPs)**. Their main difference is that IGPs provide information on reachable interior destinations to the outside world, whereas, inversely, EGPs provide information on reachable exterior destinations to the interior routers. EGPs are typically used between ISPs. Another difference between them is the **route determination**. IGPs choose the best path through the network based on some **metrics such as distance, delay, or bandwidth**. Therefore, how the metrics are used is an important design issue for IGPs. EGPs choose routes commonly according to a **routing policy**.

IGP (Interior Gateway Protocol) :-

- ✓ Within a single autonomous system
- ✓ Single network administration
- ✓ Unique routing policy
- ✓ Make best use of network resource

IGPs fall into **two categories**:

A. Distance Vector Protocols

- ✓ Routing Information Protocol (RIP V1 and RIP V2, RIPng)
- ✓ Interior Gateway Routing Protocol (IGRP)
- ✓ Enhanced Interior Gateway Routing Protocol (EIGRP)

B. Link State Protocols

- ✓ Open Shortest Path First (OSPF)
- ✓ Intermediate System to Intermediate System (IS-IS)

EGP (Exterior Gateway Protocol) :-

- ✓ Among different autonomous systems
- ✓ Independent administrative entities
- ✓ Communication between independent network infrastructures
- ✓ EGP is commonly used between hosts on the Internet to exchange routing table information.
- ✓ Examples of an EGP: Border Gateway Protocol (BGP), Exterior Gateway Protocol (Replaced by BGP)

Class full Vs Classless Routing Protocols

Class full routing protocols do not include the subnet mask with the route advertisement. Within the same network, consistency of the subnet masks is assumed. **Summary routes** are exchanged between foreign networks. E.g., of classful routing protocols: **RIP V1, IGRP**

Classless routing protocols include the subnet mask with the route advertisement. Classless routing protocols support **variable-length subnet masking (VLSM)** and **subnetting**. E.g., of classless routing protocols: **RIP V2, EIGRP, OSPF, IS-IS**.

Internet Protocol (IP)

IP is a protocol used for communicating data across a packet-switched internetwork using the Internet Protocol suite, also referred to as **TCP/IP**. IP is the primary protocol in the Internet Layer of the Internet Protocol Suite and has the task of delivering datagrams (packets) from the **source host** to the destination host solely based on **their addresses**. **For this purpose** the IP defines addressing methods and structures for datagram encapsulation. The first major version of addressing structure, now referred to as IP Version 4 (IPv4) is still the dominant protocol of the Internet, although the successor, IP Version 6 (IPv6), is **being deployed actively worldwide** (128 bits).

IPv4 Addresses

An IPv4 address is a 32-bit address. It uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet. IPv4 addresses are unique and universal. They are **unique** in the sense that **each address defines one, and only one**, connection to the Internet. Two devices on the Internet can **never have the same address at the same time**.

A protocol such as IPv4 that defines addresses has an address space. **An address space** is the total number of addresses used by the protocol. If a protocol uses **N bits** to define an address, the address space is 2^N because each bit can have two different values (0 or 1) and N bits can have 2^N values. IPv4 uses 32-bit addresses, which means that the **address space is 2^{32}** or 4,294,967,296 (more than 4 billion). This means that, theoretically, if there were no restrictions, more than 4 billion devices could be connected to the Internet. We will see shortly that the actual number is much less because of **the restrictions imposed on the addresses**.

There **are two prevalent notations** to show an IPv4 address: binary notation and dotted-decimal notation. **Binary Notation:** In binary notation, the IPv4 address is displayed as 32 bits. So it is common to hear an IPv4 address referred to as a **32-bit address** or a **4-byte address**. The following is an example of an IPv4 address in binary notation: 01110101 10010101 00011101 00000010

Dotted-Decimal Notation : To make the IPv4 address **more compact** and **easier to read**, Internet addresses are usually written **in decimal form** with a decimal point (dot) separating the bytes. The following is the dotted-decimal notation of the above address: 117.149.29.2.

Class full Addressing

IPv4 addressing, used the concept of classes. This architecture is called classful addressing. In classful addressing, the address space is divided into five classes: A, B, C, D, and E. Each class occupies some part of the address space.

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary Notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-Decimal Notation

Exercise: Find class of the following IP addresses?

1. 00000001 00001011 00001011 11101111
2. 11000001 10000011 00011011 11111111
3. 14.23.120.8
4. 252.5.15.111

Solution

1. The first bit is 0. This is a class A address.
2. The first 2 bits are 1; the third bit is 0. This is a class C address.
3. The first byte is 14; the class is A.
4. The first byte is 252; the class is E.

One problem with classful addressing is that each class is divided into a fixed number of blocks with each block having a fixed size.

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Number of Blocks for class A = 27
Number of Blocks for class B = 214
Number of Blocks for class C = 221

Block size for class A = 224
Block size for class B = 216
Block size for class C = 28

Previously, when an organization requested a block of addresses, it was granted one in class A, B, or C. Class A addresses were designed for only large organizations such as the military, government agencies, universities, and large corporations with a large number of attached hosts or routers. Class B addresses were designed for midsize organizations with tens of thousands of attached hosts or routers. Class C addresses were designed for small organizations with a small number of attached hosts or routers. A block in class A address is too large for almost any organization. This means most of the addresses in class A were wasted and were not used. A block in class B is also very large, probably too large for many of the organizations that received a class B block. A block in class C is probably too small for many organizations.

Network ID and Host ID

In classful addressing, an IP address in class A, B, or C is divided into **network ID and host ID**. These parts are of **varying lengths**, depending on the class of the address. In class A, **1 byte** defines the **network**

ID & 3 bytes define the **host ID**. In **class B**, **2 bytes** define the **network ID** & **2 bytes** define the **host ID**. In **class C**, **3 bytes** define the **network ID** & **1 byte** defines the **host ID**.

	1 st octet	2 nd octet	3 rd octet	4 th octet
Class A	Network	Host	Host	Host
	Network	Network	Host	Host
Class C	Network	Network	Network	Host

Default Mask

The length of the network ID and host ID (in bits) is predetermined in classful addressing. we can also use a **mask** (also called the **default mask**), a 32-bit number made of contiguous 1s followed by contiguous 0s. The mask can help us to find the network ID and the host ID. For example, the mask for a class A address has eight 1s, which means the first 8 bits of any address in class A define the network ID; the next 24 bits define the host ID.

Class	Binary	Dotted-Decimal	CIDR
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

IP address shortage

In the early days of the Internet, IP addresses were allocated to organizations **based on request** rather than actual need. No medium size - Hosts:

- ✓ Class A: 16 million
- ✓ Class B: 65,536
- ✓ Class C: 256

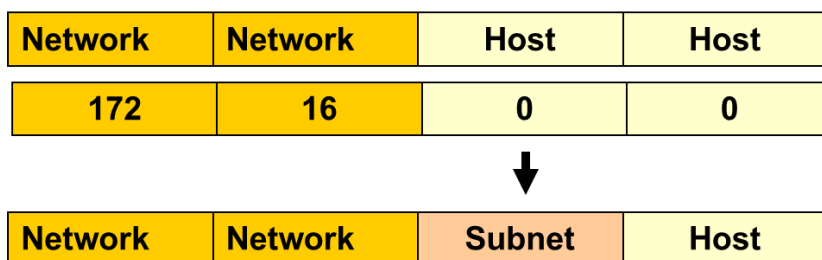
Subnet Mask

The solution to the IP address shortage was thought to be **the subnet mask**. Formalized in 1985, the subnet mask breaks a single class A, B or C **network in to smaller pieces**.

The flaws in classful addressing scheme combined with the fast growth of the Internet led to the near depletion of the available addresses. Yet the number of devices on the Internet is much less than the 2^{32} address space. We have run out of class A and B addresses, and a class C block is too small for most midsize organizations. One solution that has alleviated the problem is the idea of classless addressing.

Subnetting

During the era of **classful addressing**, subnetting was introduced. If an organization was granted a **large block in class A or B**, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called **subnets**) or, in rare cases, share part of the addresses with neighbors. **Subnetting increases the number of 1s in the mask. Subnetting is the process of borrowing bits from the HOST bits, in order to divide the larger network into small subnets.** Subnetting does **NOT** give you more hosts, but **actually costs you hosts**.



You lose **two host IP Addresses** for each subnet, and perhaps one for the subnet IP address and one for the subnet broadcast IP address. You lose the **last subnet** and all of its hosts' IP addresses as the broadcast for that subnet is the same as the broadcast for the network. **In older networks**, you would have lost the first subnet, as the subnet IP address is the same as the network IP address. (This subnet can be used in most networks.)

Classless Addressing

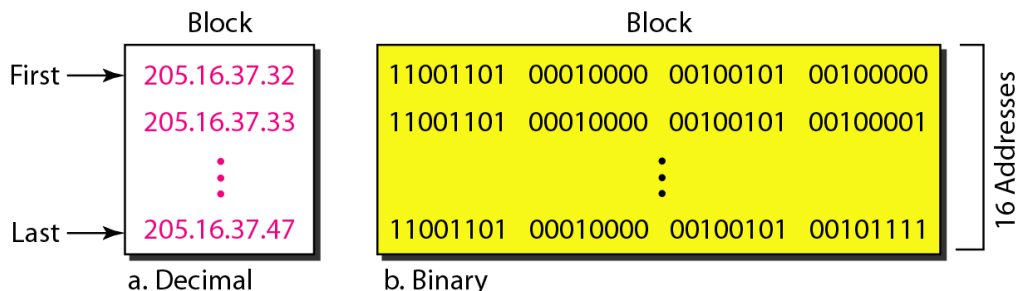
To **overcome address depletion** and give more organizations access to the Internet, **classless addressing was designed and implemented**. In this scheme, there are no classes, but the addresses are still granted in blocks.

Address Blocks

In classless addressing, when an entity, small or large, needs to be connected to the Internet, it is granted a block (range) of addresses. The size of the block (the number of addresses) varies based on the nature and size of the entity. For example, a household may be given only two addresses; a large organization may be given thousands of addresses. An ISP, as the Internet service provider, may be given thousands or hundreds of thousands based on the number of customers it may serve.

To simplify the handling of addresses, the **Internet authorities** impose three restrictions on classless address blocks:

- ✓ The addresses in a block must be contiguous, one after another.
- ✓ The number of addresses in a block must be a power of 2 (1, 2, 4, 8).
- ✓ The first address must be evenly divisible by the number of addresses.



The above figure shows a block of addresses, in both binary and dotted-decimal notation, granted to a small business that needs 16 addresses. We can see that the restrictions are applied to this block. The addresses are contiguous. The number of addresses is a power of 2 ($16 = 2^4$), and the first address is divisible by 16. The first address, when converted to a decimal number, is 3,440,387,360, which when divided by 16 results in 215,024,210.

Classless addressing Mask

A better way to define a block of addresses is to select any address in the block and the mask. As we discussed before, a mask is a 32-bit number in which the n leftmost bits are 1s and the $32 - n$ rightmost bits are 0s. However, in classless addressing the mask for a block can take any value from 0 to 32. It is very convenient to give just the value of n preceded by a slash (CIDR – Classless Inter Domain Routing notation). The address and the $/n$ notation completely define the whole block (the first address, the last address, and the number of addresses).

The first address in the block can be found by setting the $32 - n$ rightmost bits in the binary notation of the address to **0s**. **E.g. 1**, A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution:

The binary representation of the given address is

11001101 00010000 00100101 00100111 If we set **32–28** rightmost bits to 0, we get

11001101 00010000 00100101 0010**0000**

or

205.16.37.32.

The last address in the block can be found by setting the $32 - n$ rightmost bits in the binary notation of the address to 1s. **E.g.**, Find the last address for the block in above E.g., 1.

Solution:

The binary representation of the given address is

11001101 00010000 00100101 00100111 If we set $32 - 28$ rightmost bits to 1, we get

11001101 00010000 00100101 0010**1111**

or

205.16.37.47

The number of addresses in the block is the difference between the last and first address. It can easily be found using the formula 2^{32-n} . **Example** Find the number of addresses in E.g., 1.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} or 16

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number.

✓ This is particularly useful when we are writing a program to find these pieces of information.

✓ In the above example the **/28** can be represented as:

11111111 11111111 11111111 11110000 (twenty-eight 1s and four 0s).

Find

- The first address
- The last address
- The number of addresses.

- a. *The first address can be found by **ANDing** the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise*

Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

6. *The last address can be found by **ORing** the given addresses with the complement of the mask. ORing here is done bit by bit. The result of ORing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.*

Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

The number of addresses can be found by complementing the mask, interpreting it as a decimal number, and adding 1 to it.

Mask complement:	00000000	00000000	00000000	00001111
Number of addresses:	$15 + 1 = 16$			

Network Addresses

A very important concept in IP addressing is the network address. When an organization is given a block of addresses, the organization is free to allocate the addresses to the devices that need to be connected to the Internet. The first address in the class, however, is normally (not always) treated as a special address. The first address is called the network address and defines the organization network. It defines the organization itself to the rest of the world. The first address is the one that is used by routers to direct the message sent to the organization from the outside.

Address Allocation

The next issue in **classless addressing is address allocation**. How are the blocks allocated? The ultimate responsibility of address allocation is given to a global authority called the **Internet Corporation for Assigned Names and Addresses (ICANN)**. However, ICANN does not normally allocate addresses to individual organizations. It assigns a large block of addresses to an ISP. Each ISP, in turn, divides its assigned block into smaller sub blocks and grants the sub blocks to its customers. In other words, an ISP receives one large block to be distributed to its Internet users. This is called **address aggregation**: many blocks of addresses are aggregated in one block and granted to one ISP.

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- The first group has 64 customers; each needs 256 addresses.
- The second group has 128 customers; each needs 128 addresses.
- The third group has 128 customers; each needs 64 addresses.

Design the sub blocks and find out how many addresses are still available after these allocations?

Solution

Group 1

For this group, each customer needs 256 addresses. This means that 8 bits are needed to define each host. The prefix length is then $32 - 8 = 24$. The addresses are

1st Customer:	190.100.0.0/24	190.100.0.255/24
2nd Customer:	190.100.1.0/24	190.100.1.255/24
...		
64th Customer:	190.100.63.0/24	190.100.63.255/24
Total =	$64 \times 256 = 16,384$	

Group 2

For this group, each customer needs 128 addresses. This means that 7 bits are needed to define each host. The prefix length is then $32 - 7 = 25$. The addresses are

1st Customer:	190.100.64.0/25	190.100.64.127/25
2nd Customer:	190.100.64.128/25	190.100.64.255/25
...		
128th Customer:	190.100.127.128/25	190.100.127.255/25
Total =	$128 \times 128 = 16,384$	

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Group 3

For this group, each customer needs 64 addresses. This means that 6 bits are needed to each host. The prefix length is then $32 - 6 = 26$. The addresses are

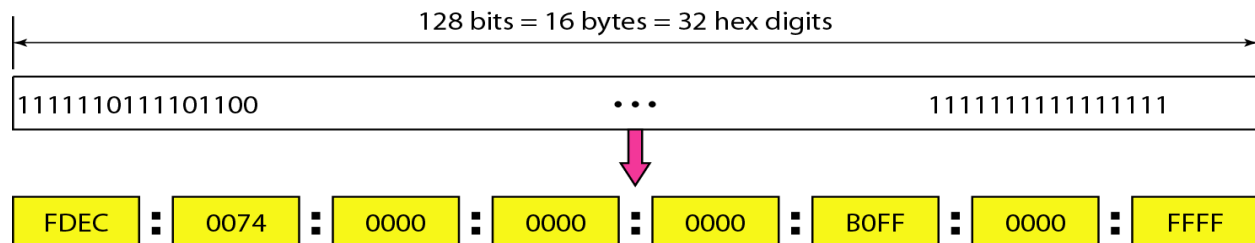
1st Customer:	190.100.128.0/26	190.100.128.63/26
2nd Customer:	190.100.128.64/26	190.100.128.127/26
...		
128th Customer:	190.100.159.192/26	190.100.159.255/26
Total =	$128 \times 64 = 8192$	

- Number of granted addresses to the ISP: 65,536
- Number of allocated addresses by the ISP: 40,960
- Number of available addresses: 24,576

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IPv6 Addresses

Internet Protocol (IP) version 4 is the most commonly used communication protocol for both the Internet and internal network environments. Although IPv4 is robust and scalable, new technologies and higher demand have paved the way for the eventual adoption of IPv6. Despite all short-term solutions, address depletion is still a long-term problem for the Internet. This and other problems in the IP protocol itself have been the motivation for IPv6. An IPv6 address is 128 bits or 32 hexadecimal digits long.



Differences between IPv4 and IPv6

Feature	IPv4	IPv6
Source and destination addresses	32 bits (4 bytes) in length	128 bits (16 bytes) in length
IPsec support	Optional	Required
Address Resolution Protocol	Broadcast ARP Request frames resolve IPv4 address to link layer address	ARP Request frames replaced with multicast Neighbor Solicitation messages
Internet Group Management Protocol	Manages local subnet group membership	IGMP replaced with MLD messages
ICMP Router Discovery	Determines IPv4 address of default gateway	Replaced with ICMPv6 Router Solicitation and Router Advertisement messages
Broadcast addresses	Sends traffic to all nodes on a subnet	Uses a link-local scope, all-nodes multicast address instead of an IPv6 broadcast address
Configuration	Configured manually or through DHCP	Does not require manual configuration or DHCP
Resource records	Uses A resource records in DNS to map host names to IPv4 addresses	Uses AAAA resource records in DNS to map host names to IPv6 addresses

Advantages of IPv6 Addressing Scheme

1. Large address space:- IPv6 uses a 128-bit address space, which allows for 3.4×10^{38} or 340,282,366,920,938,463,463,374,607,431,768,211,456 possible addresses.
2. Hierarchical addressing and routing infrastructure:- The IPv6 address space is designed to be more efficient for routers, which means that even though there are many more addresses, routers can process data much more efficiently because of address optimization.
3. Stateless and Stateful address configuration:- Stateless address configuration refers to host IP configuration without a DHCP server.

- ✓ Stateful address configuration refers to host IP configuration that uses a DHCP server.
- ✓ IPv6 supports both stateless and stateful address configuration.
- ✓ With stateless address configuration, hosts automatically configure themselves with IPv6 link-local addresses along with additional addresses advertised by local routers.
- 4. Built-in security:- IPv6 has built-in IP security, which facilitates configuration of secure network connections.
- 5. Prioritized delivery:- IPv6 contains a field in the packet that allows network devices to determine the specified rate at which the packet should be processed.
- ✓ This allows traffic prioritization or QoS. For example, when streaming video traffic, it is critical that the packets arrive in a timely manner. You can set this field to ensure that network devices determine that the packet delivery is time-sensitive.
- 6. Neighbor detection:- IPv6 uses the Neighbor Discovery protocol to manage the interaction between nodes within the same network link.
- ✓ Neighbor Discovery replaces the broadcast-based Address Resolution Protocol (ARP) with more efficient multicast and unicast communication within the same network segment.
- 7. Extensibility:- IPv6 has been designed so that it can be extended with fewer constraints than IPv4.

IPv6 Address Space

Address Syntax:

- **128-bit address in binary:**

```
0010000000000000100001101101110000000
00000000000000010111100111011
0000001010101010000000001111111111
1110001010001001110001011010
```

- **128-bit address divided into 16-bit boundaries:**

```
0010000000000001 0000110110111000
0000000000000000 0010111100111011
0000001010101010 0000000011111111
1111111000101000 1001110001011010
```

- **Each 16-bit block converted to HEX (base 16):**

```
2001:0DB8:0000:2F3B:02AA:00FF:FE28:9C5A
```

- **Further simplify by removing leading zeros:**

```
2001:DB8:0:2F3B:2AA:FF:FE28:9C5A
```

Compressing Zeros:

- **Some types of addresses can contain many zeros**

- **A contiguous sequence of 16-bit blocks set to 0 can be compressed using the double colon ::**

- **Link-local:**

```
FE80:0:0:2AA:FF:FE9A:4CA2
```

- **Can be compressed down to:**

```
FE80::2AA:FF:FE9A:4CA2
```

- **Multicast:**

```
FF02:0:0:0:0:0:0:2
```

- **Can be compressed down to:**

```
FF02::2
```

Abbreviated IPv6 addresses

Original

FDEC 0074 0000 0000 0000 B0FF 0000 FFF0

Abbreviated

FDEC 74 0 0 0 B0FF 0 FFF0

More abbreviated

FDEC 74 B0FF 0 FFF0

Gap

Example: Expand the address 0:15::1:12:1213 to its original.

Solution

We first need to align the left side of the double colon to the left of the original pattern and the right side of the double colon to the right of the original pattern to find how many 0s we need to replace the double colon.

XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX
0: 15: : 1: 12:1213

This means the original address is:

0000:0015:0000:0000:0000:0001:0012:1213

Chapter Four: Physical and Data link layer

Data Link Layer

The data link layer is responsible for carrying a frame from one hop (computer or router) to the next hop; i.e., it has local responsibility unlike the network layer. Data Link Layer Protocol Services includes

- ✓ **Packetizing** (framing or encapsulation): Encapsulate datagram/packet into frame, adding header and trailer
- ✓ **Addressing**: MAC (Media Access Control) addresses are used in frame headers to identify source and destination; different from IP addresses!
- ✓ **Error control**: Reliable delivery between adjacent nodes (error detection and correction)
- ✓ **Error detection**: Errors can be caused by signal attenuation, noise, etc. The receiver detects the presence of errors, drops the frame and waits for retransmission.
- ✓ **Error Correction**: The receiver identifies and corrects bit error(s) without resorting to retransmission. Seldom used on low bit-error link (fiber, some twisted pair). Used on wireless links: high error rates
- ✓ **Flow Control**: Pacing between adjacent sending and receiving nodes.

- ✓ **Media Access (Link) Control:** Channel access if medium is shared. Examples of link layer protocols: Ethernet, 802.11 wireless LAN, token ring, and PPP. A datagram may be carried by different protocols on the different links in the path, e.g., Ethernet on first link, frame relay on intermediate links, 802.11 on last link.

There are three categories of multiple access protocols

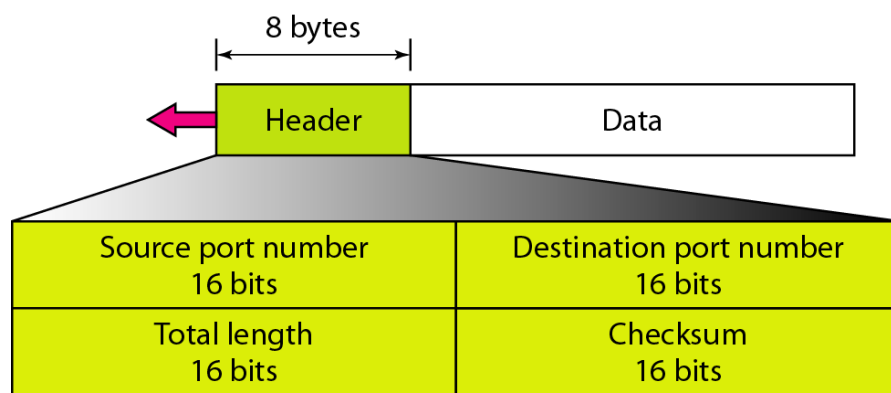
- ✓ Random Access Protocols - try your best like taxis do
 1. MA - Multiple Access
 2. CSMA - Carrier Sense MA
 3. CSMA/CD - CSMA with Collision Detection
 4. CSMA/CA - CSMA with Collision Avoidance
- ✓ Controlled-Access (Taking-turns) Protocols - get permission
 1. Reservation
 2. Polling
 3. Token Passing
- ✓ Channelization (or Channel Partitioning) Protocols - simultaneous use
 1. FDMA - Frequency-Division MA
 2. TDMA - Time-Division MA
 3. CDMA - Code-Division MA (to be covered in Chapter 4 - Wireless and Mobile Networks)

Chapter Five: Transport Layer

This layer defines how a given packet gets delivered to the appropriate process(process-to-process delivery).To allow information to be delivered to the appropriate process, we must have some way of identifying that processes.

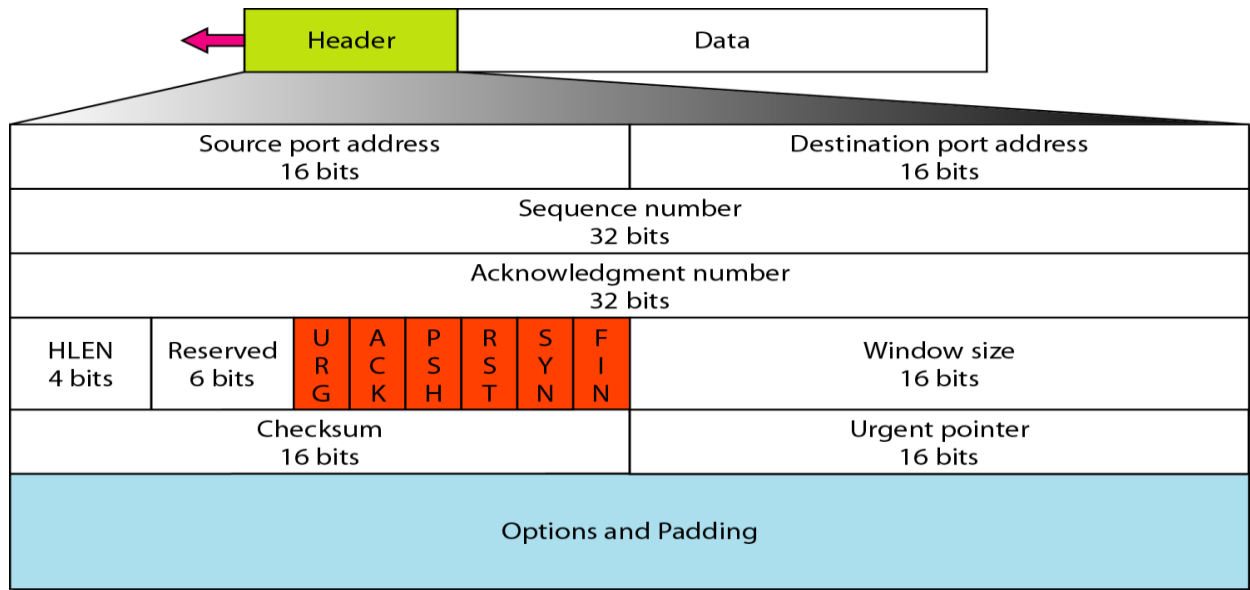
UDP User Datagram Protocol

Provides Connectionless service. It is Unreliable and mainly used for sending small amount of data. Used for Multicasting, because multicasting capabilities are embedded in UDP software. Used by a process that requires simple request-response communication with little concern for flow and error control.



TCP Transmission Control Protocol

It creates a virtual connection between two TCPs to send data. It uses flow control and error control mechanism at the transport layer level. Because the sending and receiving processes may not produce and consume data at the same speed, TCP needs buffers for storage. There are two buffers, the sending buffer and the receiving buffer, for each direction. Buffers are of hundreds or thousands of bytes, depending on implementation and not of same size. Byte locations are recycled and made available for further use.



Chapter Six: Application Layer

The Application layer, Layer seven, is the top layer of both the OSI and TCP/IP models. Provides the interface between the applications we use to communicate and the underlying network.

Application Layer Protocols

Application layer protocols are used to exchange data between programs running on the source and destination hosts.

- ✓ Domain Name Service Protocol (DNS) is used to resolve Internet names to IP addresses.
- ✓ Hypertext Transfer Protocol (HTTP) is used to transfer files that make up the Web pages of the World Wide Web.
- ✓ Simple Mail Transfer Protocol (SMTP) is used for the transfer of mail messages and attachments.
- ✓ Telnet, a terminal emulation protocol, is used to provide remote access to servers and networking devices.
- ✓ File Transfer Protocol (FTP) is used for interactive file transfer between systems.

~~~~~**End**~~~~~